



Florin Périer, Blaise Pascal's brother-in-law, climbed France's Puy-de-Dôme volcano to measure changes in atmospheric pressure at higher altitudes with a rudimentary barometer.

IN THE 1640S, FRENCH MATHEMATICIAN BLAISE PASCAL

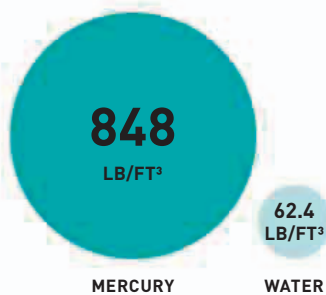
started to investigate hydraulics—the mechanical properties of liquids. He found that unlike gas, liquid cannot be compressed, so when a force is applied, it is transmitted through the liquid. Pascal's studies led to the **invention of the hydraulic press** and the **syringe**. By 1646, Pascal had confirmed Italian physicist Evangelista Torricelli's observation that a fluid would rise in a glass column because of air pressure bearing downward (see 1643–44). Pascal also predicted that this pressure would diminish at higher altitudes. He asked his brother-in-law **Florin Périer**, who lived near a mountain, to test the

idea. Receiving proof, Pascal suggested that air would thin out into a vacuum at still greater altitudes.

Polish astronomer **Johannes Hevelius's** (1611–87) greatest achievement came in 1647, when he published *Selenography* (*Description of the Moon*). The **first atlas of the Moon's surface**, it became a standard reference for years to come.

In 1648, a collection of essays written by Flemish chemist **Jan Baptist van Helmont** (1580–1644) was posthumously published by his son. Helmont had articulated an early version of the law of conservation of matter by describing a **five-year experiment in growing a willow tree**. By weighing both the plant

and soil, he had deduced that material for the tree's growth had come from water. More than a century later, experimenters found that an even greater quantity came from air, in the form of carbon dioxide.



Comparative densities
Nearly 14 times denser than water, mercury rises short measurable distances in capillary tubes, making it useful in barometers.

“WHILE [THE ATOMS] ARE MOVING... MEETING, INTERWEAVING, INTERMINGLING, UNROLLING, UNITING AND BEING **FITTED TOGETHER**, MOLECULES... ARE CREATED.”

Pierre Gassendi, French philosopher, from *Syntagma Philosophiae Epicuri*, 1649

IN 1644, FRENCH PHILOSOPHER AND MATHMETICIAN RENÉ DESCARTES (1596–1650) had described a mechanical universe that was filled with particles of matter, within which a vacuum (a space devoid of matter) was an impossibility. In 1649, French priest, experimenter, and philosopher **Pierre Gassendi** (1592–1655) rejected the notion that everything could be explained in purely mechanical terms, and proposed an alternative. He suggested that the properties of matter were determined by the **shapes of the atoms**, and that atoms joined together to make bigger molecules. Gassendi accepted the existence of vacuums and even proposed that most matter consisted of “void.” Gassendi's views anticipated later ideas concerning the bonding of atomic elements and the idea that an atom's mass is almost entirely concentrated at its nucleus (see 1911).

German physicist **Otto von Guericke** (1602–86) performed many experiments to prove that the vacuum existed. Around 1650, he invented a **piston-operated vacuum pump** with a valve system that could remove the air from a container by



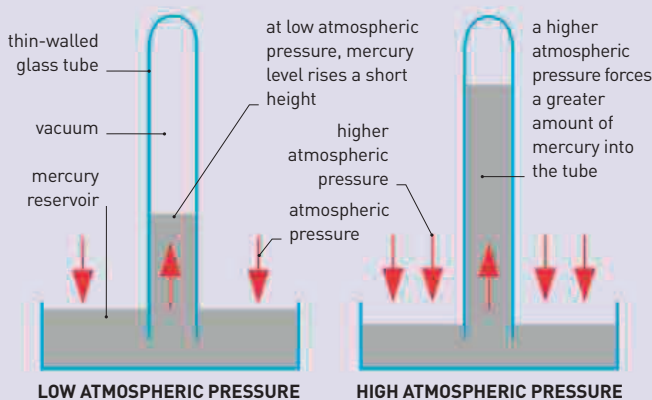
Piston-operated vacuum pump
Using an elaborate piston system, Otto von Guericke created a vacuum inside two joined hemispheres, called the hemispheres of Magdeburg, after his hometown in Germany.

“THE AIR... FLOWS ALL AROUND US. JUST AS IT **PRESSES FROM ABOVE** ON THE HEAD, IT LIKEWISE **PRESSES ON THE SOLES** OF THE FEET... AND... **ON ALL PARTS OF THE BODY** FROM ALL DIRECTIONS.”

Otto von Guericke, German physicist, from *Experimenta Nova*, 1672

MEASURING ATMOSPHERIC PRESSURE

For centuries, it was believed that air had no weight. But in fact it exerts a measurable force per surface area of the Earth. Blaise Pascal demonstrated atmospheric pressure by inverting a mercury-filled glass tube over a mercury reservoir. The tube's mercury falls to create an airless space (a vacuum), but atmospheric pressure pushes down on the reservoir to maintain a column: the bigger the pressure, the taller the column.



1645 English sailor Robert Dudley begins publishing the first atlas using Gerardus Mercator's projection of the whole known world

1645 Blaise Pascal publishes a pamphlet on his **mechanical calculator**

1646 Dutch anatomist Isbrandus van Diemerbroeck publishes a monograph on **plague**

1647 Johannes Hevelius publishes an atlas of the Moon

September 19, 1648 Florin Périer demonstrates that atmospheric pressure varies with height

1648 German naturalist George Marcgrave describes the **natural history of Brazil**

1648 Jan Baptist van Helmont's collected works on chemistry and biology are published posthumously

1649 Pierre Gassendi describes an **atomic theory of matter**

1649 René Descartes identifies the **pineal gland** as the body's control center

1650 English botanist William Howe publishes the first catalog of British plants arranged systematically

1650 Otto von Guericke devises a **vacuum pump**

1650 Pascal develops the **hydraulic press**